Vortex Lattice Method (V.L.M.)

Aerodynamic Characterization and Stall Optimization

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1. Objectives of the Project

The objectives of this project are the following:

- Apply the vortex lattice method to determine the aerodynamic characteristics of a wing.
- Optimize the stall characteristics of the wing.

2. Baseline Data

The baseline airfoil used in the study is the NACA 3217, which features, relative to the chord, a maximum camber f of 3%, a position of said maximum camber x_f at 20% of the chord ($0.20\,c$), and a maximum relative thickness t of 17%.

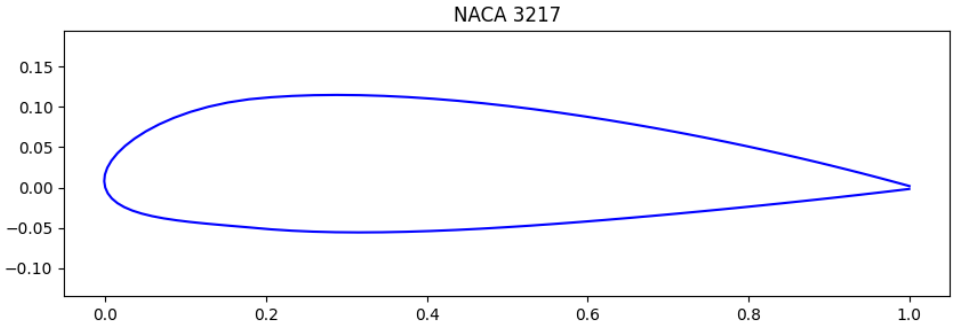


Figure 1: Geometric representation of the NACA 3217 airfoil.

The different wing parameters are obtained from the values: $W = 9, X = 3, Y = 5, Z = 5$. Resulting in the parameters shown in Table 1:

Geometric Parameters	Value
Wing Area (S_w)	$1\ \mathrm{m}^2$
Aspect Ratio (A)	15
Taper Ratio (λ)	0.65
Sweep Angle ($\Lambda_{1/4}$)	12°
Geometric Twist at the Tip (ε)	−5°
Number of Chordwise Panels (N_x)	1
Number of Semispan Panels (N_y)	10

Table 1: Geometric Parameters of the Wing.

2.1. Wing Representation

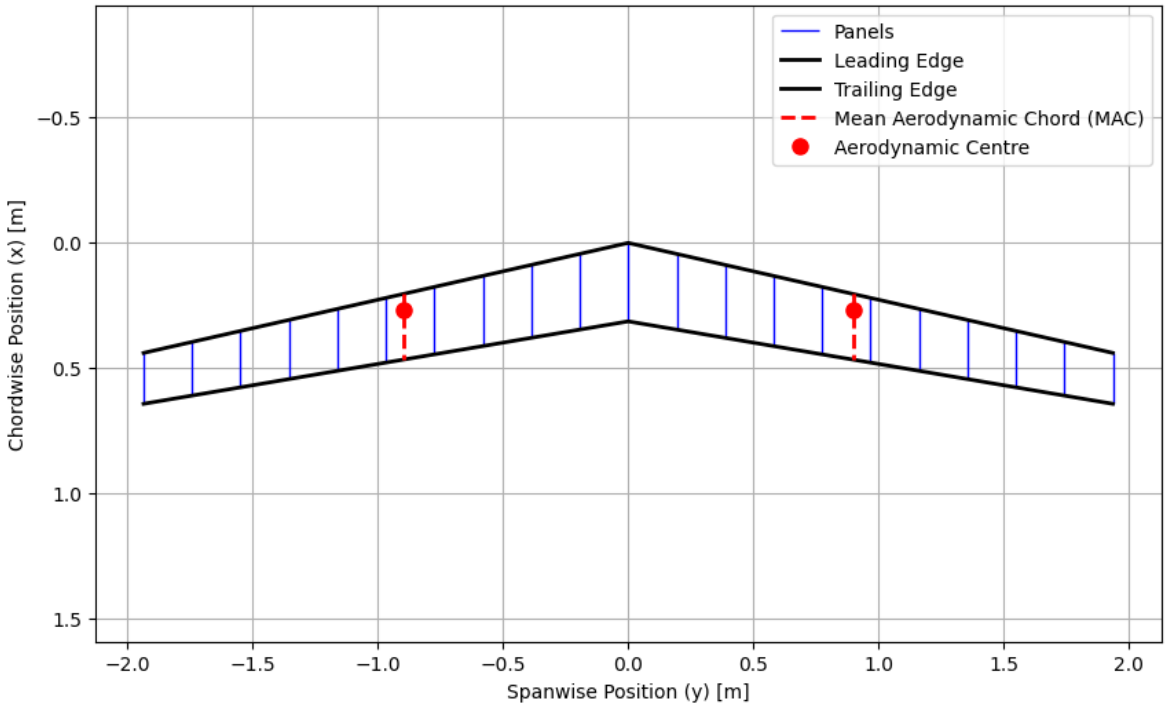


Figure 2: Wing Representation via Python Code.

3. Calculation of Different Geometric Parameters

With the data provided in the statement, other geometric parameters essential to the work can be obtained, such as[1]:

- The wing slope $\left(\frac{dy_c}{dx}\right)$ of the panel at $\frac{x}{c} = 0.75$.

Using Pistolesi's theorem and based on the NACA 3217 ($f = 0.03$ and $x_f = 0.20$). Since $\frac{x}{c} = 0.75 > 0.20$, we use the following expression (1):

$$\frac{dy_c}{dx} = \frac{2f}{(1 - x_f)^2}\left(x_f - \frac{x}{c}\right) \tag{1}$$

- The wingspan(b)

From the following equation (2):

$$A = \frac{b^2}{S} \tag{2}$$

- The root chord $(c_0)_{(y=0)}$:

Solving from:

$$S = \frac{b}{2} \cdot c_0 \cdot (1 + \lambda) \tag{3}$$

- The tip chord $(c_t)_{(y=b/2)}$:

$$c_t = c_r \cdot \lambda \tag{4}$$

- The geometric mean chord (c_{gm}):

$$c_{gm} = \frac{c_0 + c_t}{2} \tag{5}$$

- The mean aerodynamic chord (c_{am}):

$$c_{am} = \frac{2}{3} c_0 \frac{1 + \lambda + \lambda^2}{1 + \lambda} \tag{6}$$

- The aerodynamic center of the wing:

$$\tan(\Lambda_{LE}) = \tan\left(\Lambda_{1/4}\right) \cdot \frac{c_0}{2b}(1 - \lambda) \tag{7}$$

$$y_{ca} = \frac{b}{6} \cdot \frac{1 + 2\lambda}{1 + \lambda} \tag{8}$$

$$x_{ca} = y_{ca} \tan(\Lambda_{LE}) + 0.25 \cdot c_{am} \tag{9}$$

Solving the preceding expressions yields the values presented in Table 2:

Geometric params	Value
Panel slope at $x = 3c/4$	-0.0516
Wingspan (b)	3.873 [m]
Root chord (c_0)	0.313
Tip chord (c_t)	0.203
Geometric mean chord (c_{gm})	0.258
Mean aerodynamic chord (c_{am})	0.262
Aerodynamic center of the wing (x_{ca}, y_{ca})	(0.2694, 0.8994) [m]

Table 2: Other Important Wing Parameters.

4. Local Lift Coefficients

In this section, the lift linear system is solved using the vortex lattice method (VLM). For each angle of attack considered (α), the spanwise circulation distribution $\{\Gamma_j\}$ is determined by solving the system:

$$[a_{j,y}] \cdot \left\{\frac{\Gamma_y}{U_\infty}\right\} = -\{b_j\} \tag{9}$$

Where:

- $[a_{j,y}]$ represents the aerodynamic influence coefficient matrix induced by the horseshoe vortex lattice.
- $\{b_j\} = \alpha + \epsilon(y_i) - \left(\frac{dy_c}{dz}\right)_j$ represents the no-penetration boundary condition at the control point of each panel, incorporating the local geometric twist $\varepsilon(y_j)$ and the slope of the airfoil mean camber line at $x = 0.75\,c$.

Subsequently, the Kutta-Joukowski theorem is applied to obtain the local lift coefficient at each cross-section:

$$C_l(y_j) = 2 \cdot \frac{\Gamma(y_i)}{U_\infty \cdot c(y_j)} \tag{10}$$

The results are shown in the following table:

Panel	y[m]	$C_l(-12º)$	$C_l(-6º)$	$C_l(0º)$	$C_l(6º)$	$C_l(12º)$
1	0.097	-0.881	-0.342	0.198	0.737	1.277
2	0.290	-0.939	-0.379	0.181	0.742	1.302
3	0.484	-0.993	-0.420	0.154	0.727	1.300
4	0.678	-1.044	-0.463	0.119	0.701	1.283
5	0.871	-1.092	-0.505	0.081	0.667	1.254
6	1.065	-1.135	-0.547	0.040	0.628	1.216
7	1.259	-1.169	-0.585	-0.001	0.583	1.167
8	1.452	-1.186	-0.614	-0.041	0.531	1.103
9	1.646	-1.161	-0.619	-0.077	0.465	1.007
10	1.840	-0.997	-0.545	-0.094	0.358	0.809

Table 1: Lift coefficients of each panel for the different angles of attack.

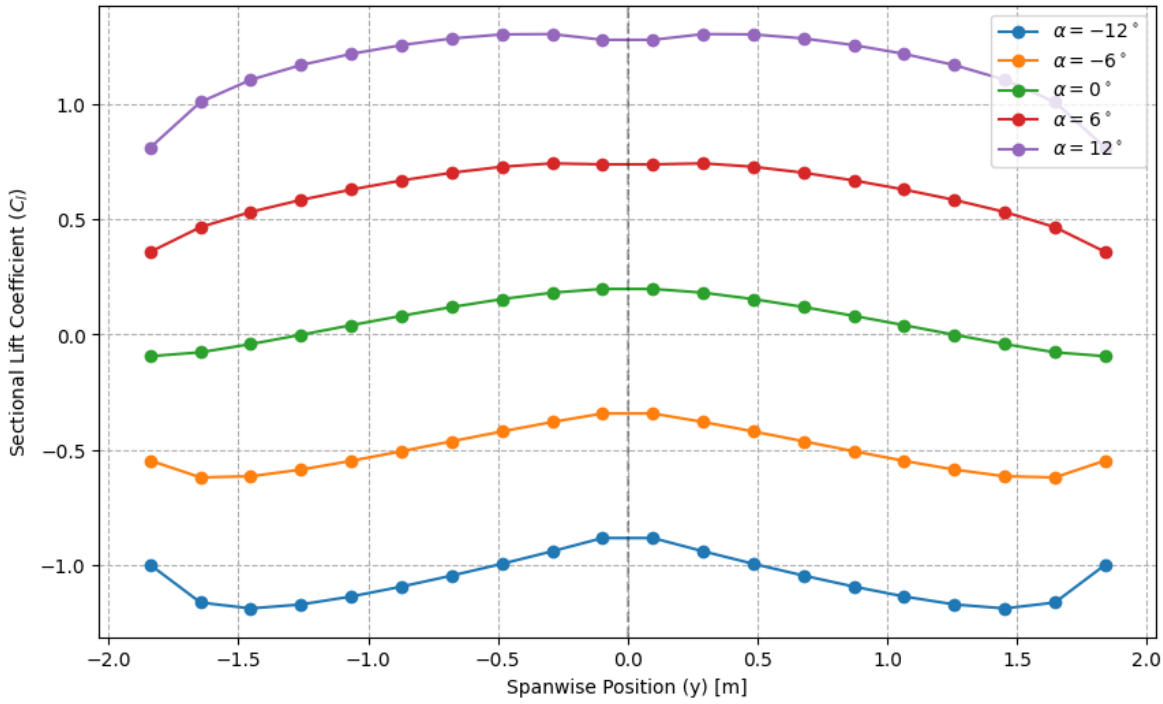


Figure 1: Local lift distribution generated with Python.

5. Global Lift Coefficient

In this section, the code calculates the global lift coefficient by integrating the local distribution along the wingspan, using the following expression (9):

$$C_L = \frac{1}{S}\sum_{i=1}^{N}(C_{l,i} \cdot c_i \cdot \Delta y_i) \tag{11}$$

To obtain the slope of the lift curve $C_{L\alpha}$, the code assumes linear behavior (valid for subsonic regimes before flow separation) with expression (10):

$$C_{L\alpha} = \frac{dC_L}{d\alpha} = \frac{C_{L,max} - C_{L,min}}{\alpha_{max} - \alpha_{min}} \tag{12}$$

Finally, it finds the value of $(C_{L,\alpha=0})$ and, using expression (11), calculates the zero-lift angle $(\alpha_{L=0})$, this value represents the "aerodynamic asymmetry" of the complete wing, which is important for the aircraft stability design.

$$\alpha_{L=0} = -\frac{C_{L,\alpha=0}}{C_{L\alpha}} \tag{13}$$

Solving the previous equations, we arrive at the following results:

$C_{L\alpha}$ [1/deg]	$C_{L,\alpha=0}$	$\alpha_{L=0}$ [º]
0.093	0.068	-0.733

Table 2: Values of the lift slope, its value when $\alpha = 0$ and the value of α when L=0.

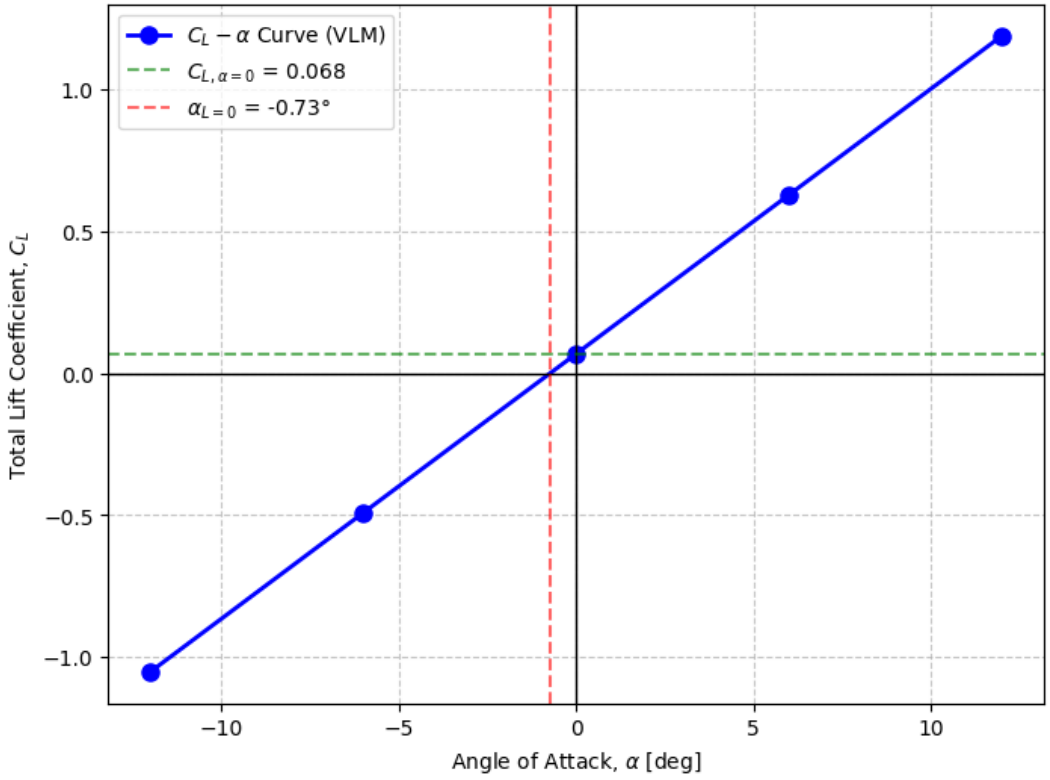


Figure 2: Representation of the (C_L vs α).

In the graph of Figure 4, a perfectly linear response is shown, highlighting the $C_{L0} = 0.068$, confirming that the airfoil camber fulfills its function of generating lift at zero incidence.

6. Basic and Unit Additional Lift Coefficient Distribution

In this section, the lift coefficient is broken down into two components: the basic one (due to the airfoil camber and twist), and the unit additional one (due to the angle of attack) for $C_L = 1$. As can be seen in expression (12):

$$C_l(y) = C_{lb}(y) + C_{la}(y) \cdot C_L \qquad (14)$$

- For the calculation of the basic lift ($\boldsymbol{C_{lb}}$), the code has solved the vortex lattice system for the zero-lift angle ($\alpha_{L=0}$). In this configuration, although the global coefficient is zero, the combination of geometric twist and airfoil camber generates a non-zero local load distribution, expressed as:

$$[A] \cdot \left\{\frac{\Gamma_b}{U_\infty}\right\} = -\left(\alpha_{L=0} + \epsilon(y) - \frac{dy_c}{dz}\right) \tag{15}$$

$$C_{l,b} = \frac{2 \cdot \Gamma_b}{U_\infty \cdot c_m} \tag{16}$$

- Now, to calculate the additional lift, the twist and camber effects have been eliminated, and the result is normalized for a unit global coefficient ($C_L = 1$):

$$C_{l,a} = \frac{C_{l,planform}(y)}{C_{L,planform}} \tag{17}$$

Solving the previous expressions, we obtain the results in Table 5:

Panel	y[m]	$C_{l,b}$	$C_{l,a}$
1	0.097	0.132	0.96
2	0.290	0.113	1.002
3	0.484	0.083	1.025
4	0.678	0.048	1.039
5	0.871	0.009	1.048
6	1.065	-0.031	1.050
7	1.259	-0.072	1.044
8	1.452	-0.111	1.023
9	1.646	-0.143	0.969
10	1.840	-0.149	0.807

Table 3: Results of the basic and additional lift coefficients for each panel.

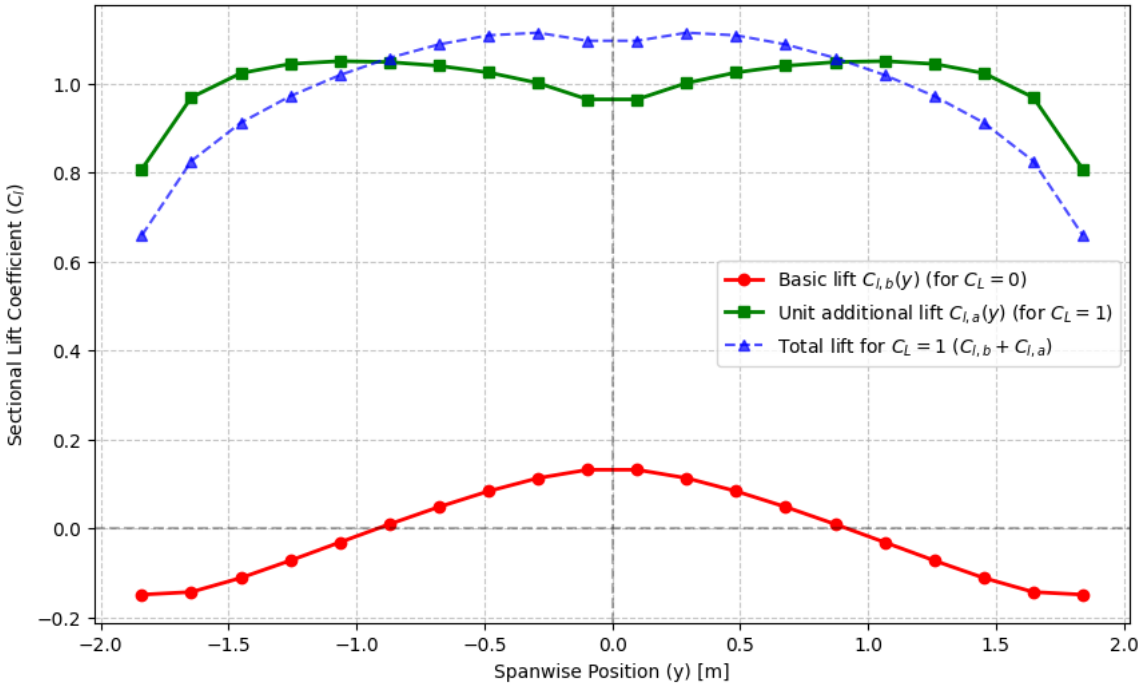


Figure 3: Representation of the basic and additional lift coefficient distributions.

7.Induced Drag Coefficients

The first step is to calculate the matrix that relates the circulation of each panel to the velocity it induces on the others, which models the downwash produced by the longitudinal wake vortices, using expression (16):

$$A_{Treffzt} = \frac{1}{2\pi} \cdot \left(\frac{1}{y_i - y_{left,j}} - \frac{1}{y_i - y_{right,j}} \right) \tag{18}$$

Then the code calculates the vertical velocity at infinity ($w_{Treffzt}$), multiplying the previous matrix by the circulation vector, and calculates the induced angle (α_i) using expression (17):

$$\alpha_i = \frac{w_{Treffzt}}{2 \cdot U_\infty} \tag{19}$$

The next step is to calculate the local drag coefficient (C_{di}) which is obtained by projecting the lift coefficient onto the direction of the inclined flow:

$$C_{di}(y) = C_l(y) \cdot \alpha_i(y) \tag{20}$$

Finally, the global coefficient (C_{Di}) is calculated by integrating these contributions over the wing surface, as seen in expression (19). The results are shown in Table 6 and Table 7:

$$C_{Di} = \frac{1}{S}\sum_{i=1}^{N}(C_{di,i} \cdot c_i \cdot \Delta y) \tag{21}$$

Panel	y[m]	$C_{di}(-12º)$	$C_{di}(-6º)$	$C_{di}(0º)$	$C_{di}(6º)$	$C_{di}(12º)$
1	0.097	0.0061	-0.0006	0.0021	0.0143	0.0360
2	0.290	0.0114	0.0008	0.0014	0.0132	0.0362
3	0.484	0.0144	0.0020	0.0008	0.0108	0.0321
4	0.678	0.0170	0.0031	0.0003	0.0087	0.0281
5	0.871	0.0199	0.0044	0.0001	0.0069	0.0249
6	1.065	0.0235	0.0060	-0.0000	0.0055	0.0226
7	1.259	0.0286	0.0080	0.0000	0.0046	0.0218
8	1.452	0.0363	0.0109	0.0002	0.0041	0.0227
9	1646	0.0487	0.0154	0.0006	0.0044	0.0266
10	1.840	0.0657	0.0213	0.0012	0.0052	0.0335

Table 4: Results of the local drag coefficient.

Angle of attack	C_{Di}
-12º	0.0251
-6º	0.0063
0º	0.0007
6º	0.0082
12º	0.0288

Table 5: Results of the global drag coefficient for the different angles of attack.

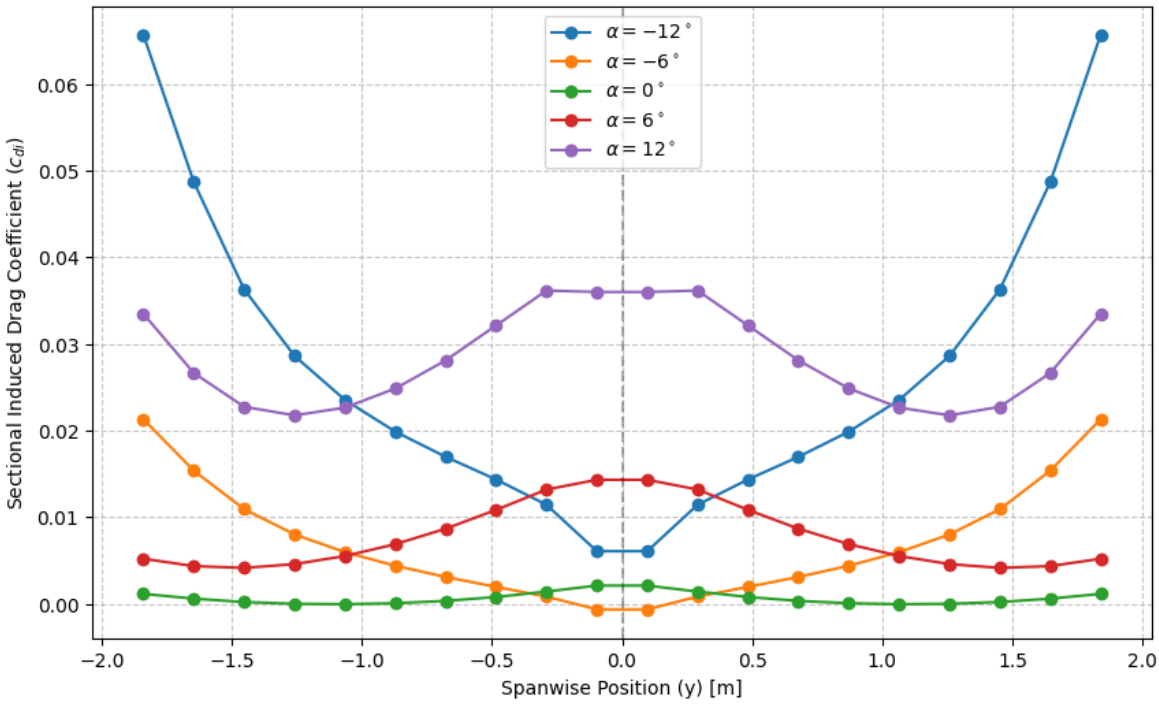


Figure 4: Representation of the local induced drag as a function of the wingspan

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8. Calculation of Pitching Moment and Longitudinal Stability

In this section, the longitudinal stability of the wing is determined by calculating the pitching moments about two reference points: the coordinate origin (OY) and the aerodynamic center of the wing x_{ca}.

The code uses the vortex lattice model (VLM); the total moment is obtained by adding the local lift contributions of each panel applied at its respective center of pressure (approximated by the line of $x_{c/4}$ in linear theory).

- First, the position of the line $x_{c/4}$ is calculated for each panel:

$$x_{1/4}(y) = |y| \cdot \tan(\Lambda_{LE}) + 0.25 \cdot c(y) \tag{22}$$

- Now the code calculates the moment of the coordinate origin with the following expression:

$$C_{M,0y} = \frac{1}{S \cdot c_{am}} \sum_{i=0}^{N} (-C_{li}(y) \cdot c_i(y) \cdot x_{1/4}(y)) \tag{23}$$

- Finally, we calculate the moment about the aerodynamic center using the moment transfer theorem (x_{ca}):

$$C_{M,ca} = C_{M,0y} + C_L \cdot \left(\frac{x_{ca}}{c_{am}}\right) \tag{24}$$

Next, in Table 8 we obtain the results.

Angle [º]	$C_{M,0y}$	$C_{M,ca}$
-12	1.115	0.034
-6	0.545	0.040
0	-0.026	0.045
6	-0.596	0.050
12	-1.166	0.055

Table 6: Pitching moment coefficients.

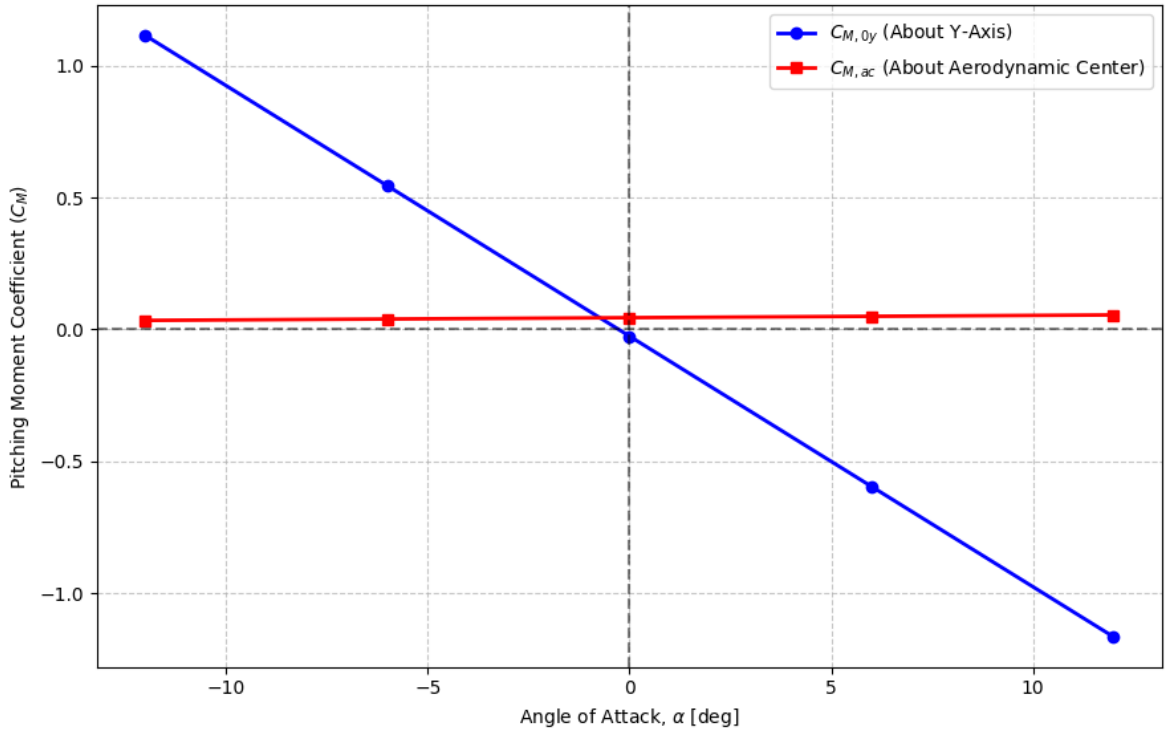


Figure 5: Representation of the pitching coefficients vs α.

In Figure 7, it can be observed that the value of the moment about the aerodynamic center is independent of the angle of attack of the wing, providing aerodynamic stability at 25% of the mean chord.

9. Sensitivity Analysis

When comparing the results obtained against XFLR5 (Figure 8), the following aerodynamic conclusions are drawn:

- **Sweep effect:** Our own VLM model faithfully reproduces the characteristic central dip in lift at the plane of symmetry caused by the sweep angle ($\Lambda_{1/4} = 12°$). This three-dimensional phenomenon is not captured by the classical Prandtl lifting-line theory (LLT) implemented by default in linear XFLR5 simulations, demonstrating the superiority of the two-dimensional VLM panel model for swept wing planforms.
- **Camber effect:** For $\alpha = 0°$ (green curve), the code rigorously incorporates the residual lift generated by the camber of the NACA 3217 airfoil ($C_{L0} = 0.068$).
- **Lift slope:** The agreement in the spacing between curves for the different angles of attack confirms the consistency of the gradient calculation $C_{L\alpha}$.

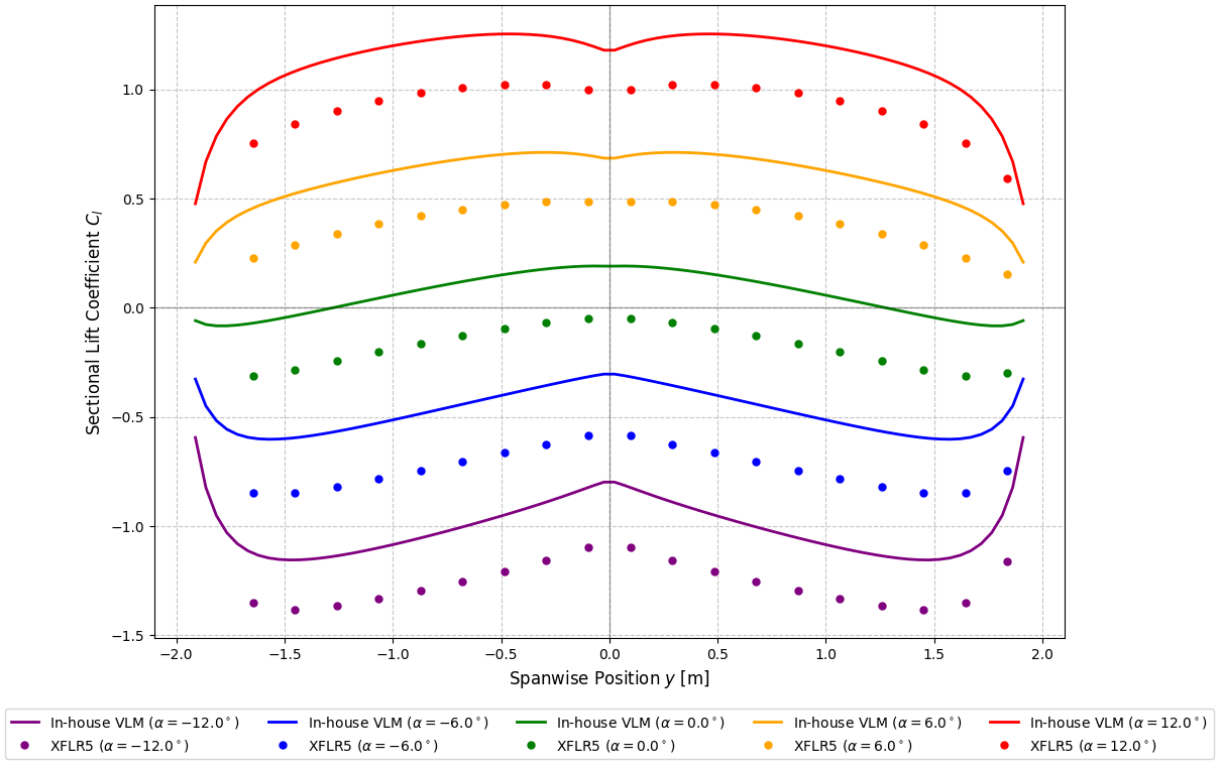


Figure 6: Comparison of local lift results calculated with code vs XFLR 5.

Now a mesh sensitivity study is carried out to determine the optimal number of panels (N_y). The results show that, from 80 panels onward, the variation in the C_L is below 0.5%, achieving robust stability.

Since the computation time increases exponentially with mesh density, ($N_y = 80$) is selected as the standard configuration, guaranteeing accuracy at a minimal computational cost.

The results are shown in Table 9:

N_y (Panels)	C_L	Time [s]
4	0.639	0.0005
10	0.634	0.0009
20	0.628	0.0031
40	0.623	0.0145
80	0.621	0.0467
120	0.619	0.1033

Table 7: Mesh sensitivity analysis ($\alpha = 6º$).

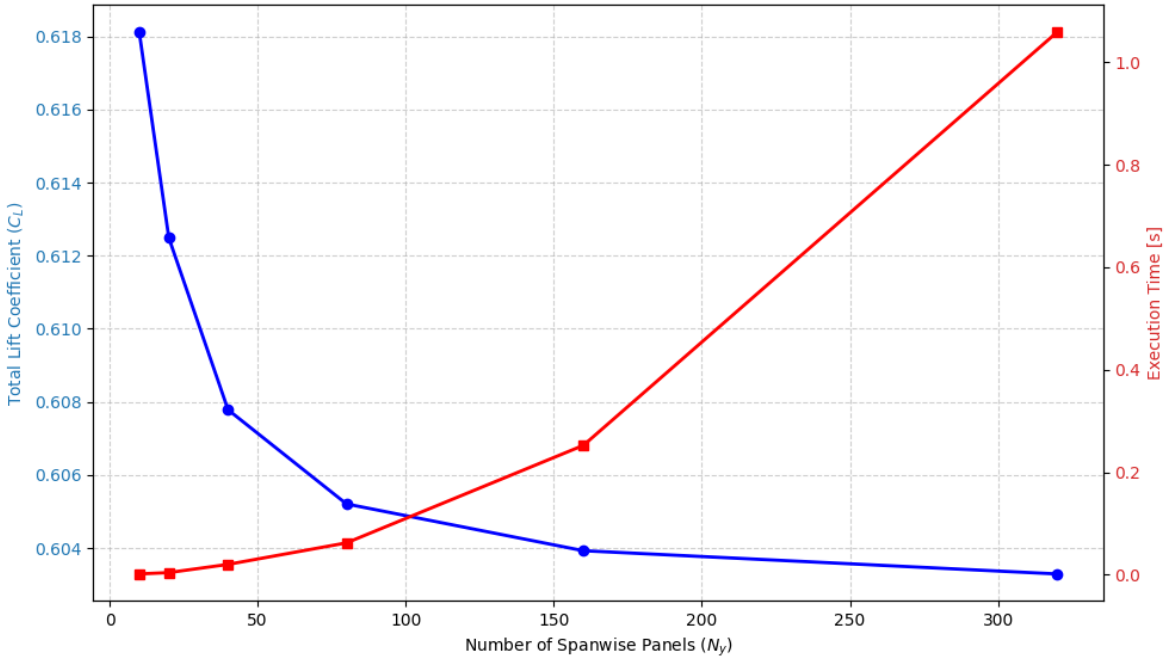


Figure 7: Aerodynamic convergence vs Computational cost.

10. Optimization of Stall Characteristics

The objective of this section is the optimization of the initial wing design, with the parameters calculated in section 2, towards a wing model that guarantees that the loss of lift occurs at a point less than or equal to 40% of the wing (where $y = 0$ is the wing root attachment point at the fuselage) and that the slope of this lift as a function of y is at least $\frac{dC_l}{dy} = -0.2$.

Methodological note: In the present analysis, under the hypothesis of a linear potential regime, the criterion adopted for stall onset is the position of maximum local lift demand ($C_l(y)$). It is assumed that the airfoil maintains a maximum capacity $c_{l,\max}$ that is uniform along the wingspan, so that the point where the lift peak is reached coincides with the critical section where the boundary layer will separate in real flight.

To achieve these objectives, the code iterates by varying the different wing parameters: the aspect ratio (AR), the taper ratio (λ), the wing twist (ε), the sweep angle (Λ) and the airfoil, that is, the NACA.

The most important parameter for the required optimization is the geometric twist (ε). By increasing the negative twist, local angle of attack is subtracted at the wing tips, shifting the maximum peak of the lift coefficient (C_l) directly towards the wing root, thereby bringing the stall point forward. The penalty of this increase is suffered mainly in the cruise phase since, due to the high twist, the tips could end up generating negative lift, which sharply increases parasitic drag and generates stresses on the wing.

Another very important parameter is the taper ratio (λ). Increasing the taper ratio allows us to obtain an increasingly rectangular wing, which provides more physical chord to distribute the load and avoid premature stall. However, this increase in taper ratio shifts the position of the center of pressure outwards, causing the bending stresses on the wing to be greater. Furthermore, by losing the elliptical distribution, the wing loses aerodynamic efficiency.

As for the aspect ratio, it barely influences the stall onset stage. The problem with leaving such a short and wide wing is that the induced drag skyrockets. At low speeds the aircraft will be safe and viable, but on cruise the wingtip vortices will heavily penalize fuel consumption.

On the other hand, there is the airfoil. Although the VLM does not notice the change of airfoil when calculating the load demand, in real life it is what sets the limit before the boundary layer separates. Placing a thick airfoil at the root ensures that the stall begins there smoothly and warns you with vibrations before dropping. The drawback of changing to a thicker airfoil is that it generates much more drag, negatively affecting the aerodynamic efficiency of the wing.

The algorithm operates by iterating values and creating a solution space that meets the established objectives. After finding all viable combinations of Aspect Ratio (by modifying the wingspan), Sweep angle, Taper ratio and geometric Twist, the code prioritizes the combinations that change the parameters the least with respect to the original ones, obtaining the five best possible combinations.

The best combination obtained by the code is the one with the values of:

NACA 4417, $AR = 7.24$, $\Lambda = 14.74°$, $\lambda = 0.905$, $\varepsilon = -9°$,

Leaving a stall stage at 15% of the semispan with a gradient of C_l as a function of y, $\frac{dC_l}{dy} = -0.201$.

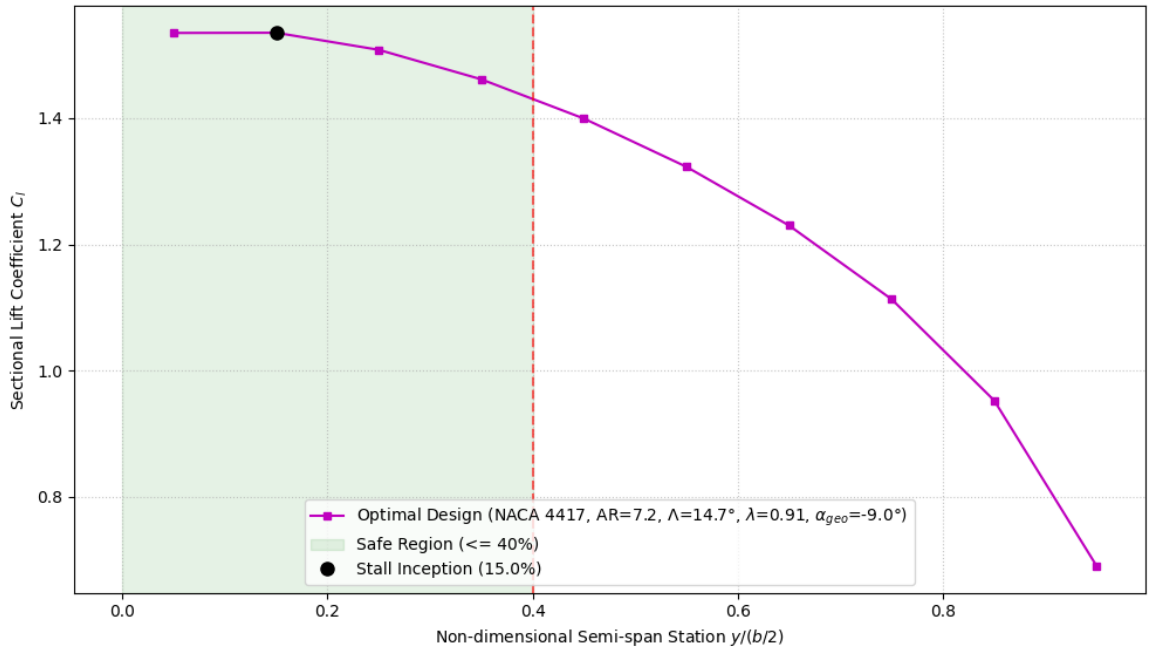


Figure 8: Lift Distribution on the Wing and Stall Point of the Optimized Wing.

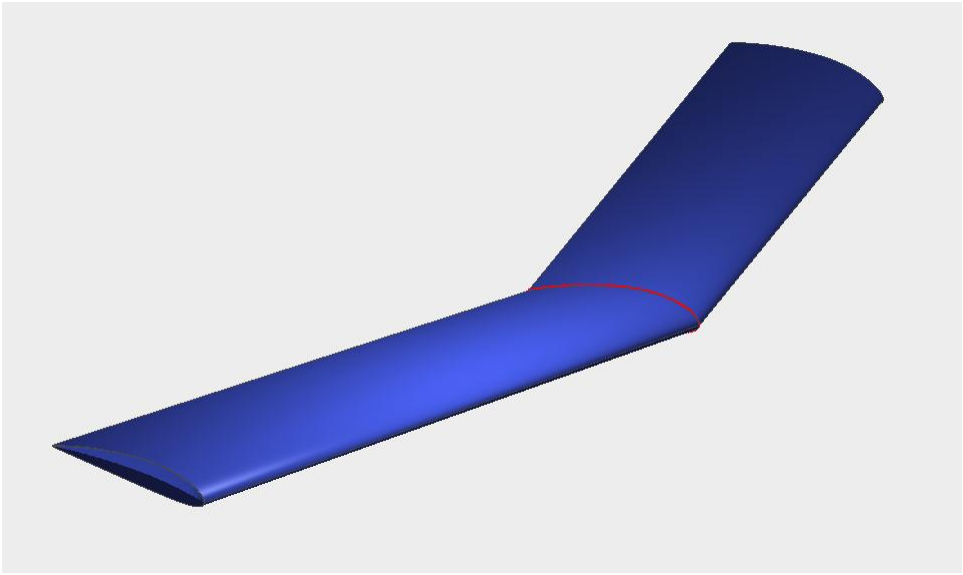


Figure 9: Representation of the Optimized Wing. Software used: XFLR 5. [2]

11. Conclusions

In this document, a numerical model based on the vortex lattice method (VLM) has been developed and validated for the aerodynamic characterization of a three-dimensional wing. In the first phase, the fundamental geometric parameters were defined, including the equivalent chords and the exact location of the aerodynamic center. Subsequently, the local distributions and the global lift coefficient were calculated, breaking the latter down into its basic and additional components to understand the isolated effect of geometry and incidence. Likewise, the induced drag was quantified, and the longitudinal stability was evaluated through the pitching moments.

The robustness of the results was favorably compared against the commercial software XFLR5 and supported by an empirical mesh convergence study. As the culmination of the project, the predictive capability of the code made it possible to optimize the baseline geometry, iterating on parameters such as twist and taper ratio to achieve a final design that safely delays and controls the onset of stall.

12. Appendix

Code and entire project on GitHub: Link.

Code in Google Colab: Link.

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13. References

[1] Gandía, Fernando, «Aerodinámica-Cap. VII- Alas en régimen incompresible»,
[2] Deperrois, André, *XFLR 5*.